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ARTICLE

Road to Resettlement: Understanding Post-disaster Relocation and Resettlement Challenges and Complexities Through a Serious Game

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Abstract

Resettlement and relocation are among the most difficult policies to put into practice, but they may be the best ways to minimize future risks to settlements exposed to natural hazards both before and after disaster events. As climate-related disasters and forced migration become increasingly common worldwide, governments, humanitarian or development actors, and policymakers must now prioritize the implementation of a dignified and effective resettlement program as part of their planning and management responsibilities. Much of this effectiveness depends on the stakeholders and beneficiaries' understanding and knowledge of the different resettlement phases, culture and customs of affected populations, activities, and the associated implementation challenges, costs, and benefits. Serious games are used in a variety of contexts to increase awareness, train and build capacity in stakeholders and beneficiaries. This article presents a serious game developed to educate practitioners, local agencies, students, and the public to understand the complexities and challenges involved in a successful resettlement. The game is based on a real proposed resettlement project initiated in the Chiradzulu District in southern Malawi after Cyclone Freddy in March 2023, which caused widespread flooding and landslides, forcing some villages to relocate permanently. The progression in the Road to Resettlement Game consists of six primary levels: land and site preparation, housing and livelihood, water, sanitation, and hygiene, health, education, and protection. These levels are meant to be completed in a sequence that adheres to the principles of resettlement. By engaging in the serious table-top board game, players gain an understanding of the resettlement activities, their sequence, and the associated practical (technical and social) and financial challenges.

Keywords Cyclone freddy · Displacement · Malawi · Post disaster recovery · Resettlement · Serious game

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1 Introduction

Globally the total number of disasters has seen a relative increase of 214% since 1970. The International Disaster Database (EM-DAT) has reported over 26,000 major disasters worldwide since 1900 (Ritchie et al. 2023). Further analysis of these records revealed that, in contrast to geological disasters (such as earthquakes), hydrological disasters—like floods—exhibited an upward trend. While the death toll from geological disasters like earthquakes is higher, hydroclimatic disasters like floods and storms affect a greater proportion of the population (Correa et al. 2011). According to the IPCC report, in some of the world's most vulnerable areas, climate change has increased the frequency and intensity (or magnitude) of hydroclimatic extreme events like floods, droughts, wildfires, and extreme heat events and this trend is predicted to continue (Trisos et al.

2022). For instance, certain African nations are vulnerable to multi-hazard events that could greatly increase losses to the economy, health, and environment (CRED 2023; IPCC 2023a; IPCC 2023b). Floods are occurring more frequently in Africa, which impacted up to seven million people in 2020 alone. Compared to the previous 20 years, flood events have increased six times more than drought events (+ 180%) in Africa and have had a greater overall impact on the continent (CRED 2023).

Disaster events (both natural origin and human made) cause significant number of forced temporary (evacuation) and permanent internal and international displacement (that is, refugees and asylum seekers) (World Economic Forum 2020; UNFCC 2022; IDMC 2023). Estimates show about 200 million people were displaced due to natural hazard related disasters between 2008 and 2015, with floods accounting for nearly 64% of these cases (Jamshed et al. 2019). While many of the displaced population after disasters can return to their original locations, considerable number of them are not able to do so due to the existence of hazard sources. These people are permanently relocated and are resettled either in the existing settlements or in new settlements. Permanent post-disaster displacement in new settlements has been very complex and challenging for both the affected people and policymakers (Badri et al. 2006). A successful resettlement (when necessary) requires careful planning and design, effective partnership and trust among the stakeholders, sufficient financial and human resources, and political supports. Knowledge of the upcoming challenges and complexities and examining possible solutions can greatly contribute to the success of resettlement initiatives. Serious games can be used to enhance knowledge and promote collaboration and trust building among the stakeholders.

During the last decade, gamification—activities that mimic game-like experiences—has drawn a lot of attention, particularly in the area of climate change engagement, where games have been shown to be highly effective in behaviorally, affectively, and cognitively connecting a variety of players with a problem (Fernández Galeote et al. 2022). Serious games (SG) have been recognized as an effective tool for knowledge creation and translation, particularly in bridging the gap between scientific research and action in addressing climate change (Lawrence and Haasnoot 2017; Flood et al. 2018). As a novel learning platform, SGs allow diverse groups of participants to take on different stakeholder roles, fostering collective and collaborative problem-solving for complex social, environmental, and health issues (Yu 2019). Game-based learning demonstrates a notably higher level of effectiveness and a longer duration of participant engagement when compared to non-game-based learning. Game-based learning exhibits significantly higher levels of motivation, desirability, and helpfulness for both learners and educators (Yu 2019). Empirical evidences indicate that learners who play serious games demonstrate significantly higher performance and skill development (Roozeboom et al. 2015; Min et al. 2022). Therefore, use of serious games is increasing, particularly in understanding complex social issues and decision and policy making including disaster and emergency management.

The main goal of this article is to present and introduce a serious game named "Road to Resettlement" developed based on the post-disaster relocation and resettlement of marginalized, under-resourced, and under-connected communities in Malawi. Developed for a variety of users, the game's main objectives are to addresses four key issues in post-disaster recovery processes: (1) to enhance participation and cooperation among different stakeholders, including government, nonprofits, and affected and/or at-risk communities; (2) to educate various stakeholders about the specific challenges and complexities of post-disaster resettlement; (3) to identify potential risks and barriers as well as the chance to modify resettlement programs before they go into effect; and (4) to provide an educational and training tool for future professionals and those involved in post-disaster resettlement. The game's design allows it to be quickly customized to meet context-specific requirements. It also offers a comprehensive list of essential actions that are necessary during the chaotic resettlement period, making it a valuable tool for emergency managers. Other contributions include giving marginalized voices a forum to be pushed to the forefront of the issue and helping to establish a trust-building atmosphere early on among the key stakeholders.

2 Background and Literature Review

Relocation and resettlement in the context of disaster management has been studied extensively from different dimensions and perspectives. Balachandran et al. (2022) compiled more than 500 of such studies and classified them into three main streams. Significant number of studies have examined relocation and resettlement during post-disaster recovery and reconstruction. The majority of these studies are case studies examining post-disaster relocation and resettlement from social, psychological, planning, and engineering aspects. They provide insights into factors that contribute to the failure and success of post-disaster resettlement and sometimes offer guidelines for planning effective relocation (World Bank 2004; Jha et al. 2010; Correa et al. 2011; Correa 2011; UNHCR and Brookings Institution 2014; UNHCR et al. 2017). Overall, it is argued that post-disaster situation can provide opportunities to mitigate risks by relocating people from hazard areas and create opportunities for further development (Badri et al. 2006; Imura and Shaw 2009). The second stream includes diverse number of studies that

focus on all forms of pre-disaster relocation and resettlement (also referred to as pre-emptive relocation) aiming to reduce potential risks caused by natural, technological, and human-made hazards, climate change (Petz 2015), and development projects such as dams and industrial facilities (Cernea 1988). Because the majority of such resettlements are involuntary, they are even more complicated than the post-disaster relocations. Considerable number of studies of relocation and resettlement before and after disasters are critical of these approaches and argue that relocation and resettlement of people make them more vulnerable and marginalized and deprive them of their identity and place attachment (Jenkins 2016; Simms et al. 2021). Since the Road to Resettlement serious game focuses on post-disaster relocation and resettlement, this review will not cover proactive relocation and resettlement.

In developing the Road to Resettlement Game, we focused on pos-disaster relocation and resettlement. Of the four pillars or phases of disaster management (mitigation, preparedness, response, and recovery), post-disaster recovery is the most complicated one due to its extensive scope, depth, and need to balance considerations with urgency. While discussions with victims and communities can go on for months or even years, recovery speed is crucial for maintaining businesses, public services, and adequate housing. In case of major disasters, the recovery process may take several years or decades. Post-disaster resettlement or relocation is a strategy in the post-disaster context that aims to move impacted individuals to safe areas when the original land is no longer suitable for human habitation due to either the site's disappearance or the existence of risk conditions that cannot be reduced through alternative means (Jamshed et al. 2019; Chen and Tsai 2021). Resettlement starts to gain interest as a disaster risk reduction measure, especially in scenarios where all other options for mitigating the risk are neither applicable nor effective (Correa et al. 2011). Although very few studies have so far examined planned or preventative resettlement, it is increasingly gaining favor as a long-term disaster management strategy due to its comprehensive plan for resettlement by evaluating potential losses to physical infrastructure, livelihood demands, and emotional and cultural attachments to a place (Chen and Tsai 2021).

Permanent resettlement or relocation is a challenging process because it involves uprooting an entire community from a setting where society has developed and adapted over centuries and relocating them to a new place could have worse effects than the disaster itself. These adjustments, which can be social, economic, political, or a combination of the three, shape the individual and societal realities. When these realities are disturbed, it can lead to serious physical and cultural crises (Oliver-Smith 1991). These crises may include social fragmentation, political marginalization, loss of privacy, landlessness, and unemployment (Birkmann et al. 2013; Jamshed et al. 2019). Most commonly, resettlement sites are not of the same size and the quality of resettlement site does not match the original living areas.

Resettlement is also a multidimensional process, which if not done correctly and thoroughly may lead to higher negative impacts than the disaster itself (Correa et al. 2011). The impoverishment risks and reconstruction (IRR) model also identified eight primary risks associated with the measure: Lack of access to common property assets, food insecurity, unemployment, homelessness, marginalization, landlessness, morbidity, and social disarticulation are some of the issues that accompany these conditions (Cernea 2023; Chen and Tsai 2021). However, it is argued that resettlement can also lead to positive development outcomes by bringing communities closer to social, economic, and infrastructural services (Badri et al. 2006; Jamshed et al. 2019). To achieve positive outcomes from resettlement and relocation, it is critical for the planning and managing process to aim for development and long-term sustainability (Arnall 2019). Although resettlement aims to do both—limit exposure and provide social and economic opportunities to affected populations, without careful planning and management it can lead to new vulnerabilities and risks (Correa et al. 2011; Jamshed et al. 2019). Unrealistic timeframes, a lack of community involvement, and a poor comprehension of the social connections already in place are examples of planning process failures (Chen and Tsai 2021). Resistance from the affected population to leave the original site or abandon the new site after relocating to repopulate the old site has also been a common problem in many post-disaster resettlement programs. Such refusal to leave an affected site despite its utter destruction and/or dangerous living conditions has been explained as non-rational human behavior, caused by their deep connections with the land, culture and traditional coping mechanisms, loss of material belongings, and the success or failure of resettlement programs themselves due to their complex and lengthy adaptation process (Oliver-Smith 1991).

Both voluntary and involuntary post-disaster resettlements are significantly more complicated than traditional development resettlement, particularly since their primary goal should be to increase community resilience (Correa et al. 2011; Jamshed et al. 2019). Beyond the reconstruction of physical infrastructure like housing, post-disaster resettlement must also address social capital, the restoration of livelihoods, and mental support. For instance, home is an important component of resettlement; however, if home is not viewed as more than just a place to live but also as a community and creating a sense of belonging, the process of resettlement may result in maladaptation. It is important to consider how the victims' livelihood, education, day-to-day activities, and chances to develop social capital will be affected by a new location. The needs of the impacted

community should be the primary consideration when choosing a resettlement site (Chen and Tsai 2021).

With increasing floods and cyclones, several countries are adopting post-flood resettlement as a recovery measure for their populations (Jamshed et al. 2018). Model villages (MVs), which Pakistan employed to relocate incredibly vulnerable populations following the 2010 floods, are one such instance. To provide flood-affected populations with better living conditions, MVs are designed to be resilient villages with the most basic modern amenities possible, including housing, electricity, sanitation, water supply, health and education facilities, veterinary services, and increased livelihood opportunities. Two significant discoveries were made by Jamshed et al. (2019) during their assessment of the Pakistan MVs: (1) As with other post-disaster resettlement strategies, MVs were also found to prioritize environmental and infrastructure factors over socioeconomic considerations; and (2) NGOs outperformed the public sector in executing the resettlement program by allocating funds for livelihood options, skill development, facility upkeep and operations, and educational opportunities.

Arnall's (2019) research indicated that in order for national level resettlement programs to be successful, they must take into account long-standing government perceptions of rural populations, maintain conditions that allow affected populations to retain their individuality and agency, and pay close attention to how those affected internalize the stories and messages created and passed along during relocation. According to Xu et al.'s (2022) study on resettlement and household adaptive capacity in China, relocated households need to be supported through training, skill development, and livelihood diversification to improve their livelihood resilience and adaptive capacity both in the short and long term for post-disaster resettlement to be successful. Through a shared vision and post-disaster resettlement, Alaniz's work in Honduras has shown that creating a strong sense of community culture is essential to achieving positive results. The application of empowerment strategies that ignore short-, mid-, and long-term recovery actions and lack organizational support in addition to technical assistance also impedes the achievement of positive community development outcomes (Alaniz 2017; Imperiale 2019). Resettlement should be carried out in incremental, participatory steps with the goal of giving displaced communities a sense of security, comfort, and ownership (Jamshed et al. 2019).

Despite research demonstrating that citizen participation is essential to the long-term viability of the solution, it is frequently absent from most resettlement programs. For instance, Jacobs and Almeida (2021) discovered that while the government's 2019 resettlement program in Mozambique after Cyclone Idai was successful from a short-term humanitarian standpoint, the program's long-term objectives were lost because of citizens' lack of involvement in the decision-making process and inadequate communication efforts. Among the problems that cause missed engagement opportunities were unclear information about the conditions of the resettlement site, insufficient details about the voluntary nature of resettlement, and the absence of a formal consultation process regarding site selection. Families became unprepared, reluctant to relocate, and disempowered because of the scarcity of employment opportunities. Additionally, the authors discovered during their fieldwork that some of the resettled families were already returning to the city (the original site) because they were accustomed to a more developed lifestyle than their new rural living conditions, and the resettlement site lacked basic living conditions because most of the infrastructure was still under construction. The key findings of Jacob and Almeida's (2021) research in Mozambique show that communication with and participation of local communities, along with collaborations between short-term humanitarian aid providers and long-term development organizations, are critical to the sustainability of the resettlement program and the development of a sense of ownership over the new location. Therefore, creating educational, knowledge sharing, and community engagement tools that can be used to communicate some of the key challenges, advantages, and disadvantages as well as key activities that need to be taken for a successful and dignified resettlement is very important.

Finally, building based on lessons learned from "migration with dignity" (McClain et al. 2020) and post-disaster relocation and resettlement studies and practices, a new direction is emerging around safe and dignified resettlement. This framework can address major resettlement issues and challenges. Safe and dignified resettlement aims to maintain cultural integrity and ensure access to livelihood opportunity, education, employment, social security and safety, and health care without losing the conditions that displaced population had in their original settlement (Farquhar 2015). Like migration with dignity, safe and dignified resettlement is a disaster recovery measure with roots in human and sociocultural rights, secure livelihood, gender balance, equity, inclusivity, autonomy and choice, and engagement in civic and sociopolitical aspects (Daly and May 2018).

Serious games design has emerged as a cutting-edge approach to thoughtfully integrate various community members with disaster risk management (DRM) objectives. For example, Goretti and Musacchi (2024) developed three digital serious games for encouraging youth participation in seismic risk management by integrating these games into civil protection activities such as understanding seismic phenomena to implementing prevention and preparedness measures by students and citizens. Another excellent example of this is the Levee Dilemma Game, which Chan and Liao (2023) used to study decision-making thought processes in challenging circumstances. They involved 228 participants

in an interactive computer game to determine how social cues, communication, and climate uncertainty influenced players' decisions to build levees (Chan and Liao 2023). A third example of this innovative approach is the application of Costa Resiliente Serious Game on the resilience skills of a coastal community in southern Chile by Villagra et al. (2023), which revealed that such games may have positive effects at people's cognitive, emotional, and behavioral levels.

Many other serious games have become popular in recent years in environmental protection (Madani et al. 2017; Georgiou et al. 2023; Tan and Nurul-Asna 2023), climate change adaptation (Fernández Galeote et al. 2022; Razali et al. 2022), and migration and refugee studies (Gabriel 2021), but their use in resettlement context is still very limited. While many risk mitigation or adaptation measures include relocation or resettlement as a possibility, the specifics of the relocation process are rarely discussed (Balachandran et al. 2022). In research related to resettlement, two notable and somewhat related examples of such games are those developed by Beirão et al. (2016) and Jackson et al. (2020). Beirão et al. developed a computer simulation based serious game for social housing relocation by integrating lessons learned from several test scenarios, and Jackson et al. employed a serious game that immerses players into a journey of reintegrating ex-offenders into society after they are released from jails—a scenario that differs greatly from post-disaster resettlement.

3 Methods

We developed the Road to Resettlement serious game based on a real proposed resettlement project in southern Malawi following Cyclone Freddy in March 2023. We applied the safe and dignified resettlement framework to develop the game. In this section, the resettlement case and the safe and dignified resettlement framework are presented.

3.1 Resettlement Case

Cyclone Freddy brought strong winds, mudslides, and landslides into 15 districts in southern Malawi from 12 to 15 March 2023, triggering widespread human, property, infrastructure, and environmental impacts. The cyclone caused a significant number of fatalities (1216) and injuries (1332). It also caused widespread displacement (659,278), forcing people to stay in more than 747 camps and schools (Adriano et al. 2023). Due to the ongoing flooding and the possibility of future landslides, many of the displaced families were not able to return to their original settlement. This situation prompted governmental and nongovernmental actors to consider relocation and resettlement as viable options. Many humanitarian agencies responding to the post-disaster recovery also voiced their support for the government to relocate more than 100,000 people through participation in joint feasibility assessments, in districts where land has been identified for relocation (OCHA 2023).

Public health issues have been among the key concerns before, during, and after such large disaster events in Malawi. When Cyclone Freddy made landfall, the country was already dealing with a cholera outbreak (Lutala and Makwero 2023). Significant damage to health facilities and road networks and population displacement made the situation even more complicated. Affected communities were forced to use single water sources during the displacement period, which increased disease transmission risks—in this case, cholera, endemic in Malawi since 1998 (World Health Organization 2023). Heavy rains exacerbated the already fragile health system facilities and infrastructure, providing perfect conditions for the deadliest cholera outbreak in the country (Adriano et al. 2023).

Malawi's geographical location makes it vulnerable to extreme weather events (UNICEF 2019), and because of climate change, the frequency of these events and ensuing secondary hazards are rising. In January 2022, tropical storm Ana caused heavy flooding and landslides in about 19 districts. The districts of Phalombe, Mulanje, Chikwawa, and Nsanje were particularly affected and more than 190,429 people were displaced (IFRC 2022). As a result, resettlement has been considered a potential solution for those communities that are highly vulnerable to such adverse events.

This case study focussed on Chiradzulu District, which experienced a high number of landslides during Cyclone Freddy. Figure 1 shows the landslide hazard in southern Malawi and the case study relocation village. Our research indicated that, of 14,601 villages in southern Malawi, 267 are in areas with a very high risk of landslides (shown as red on the maps), and 967 are in high-risk areas (shown as orange).

Mtauchira Village located in Chiradzulu District was among the communities that was significantly damages by a landslide during the cyclone (Fig. 2). A total of 59 houses were damaged or destroyed and about 300 people were made homeless and displaced. The landslide originated from the steep, deforested slope of Chambe Peak and turned into a huge mudslide that destroyed much of the Mtauchira Village instantly.

3.2 Safe and Dignified Resettlement Approach

Resettlement with dignity approach has been applied in developing the resettlement plan for Mtauchira Village that is one of the numerous villages being considered for relocation and resettlement purposes (Fig. 3). As part of this process, a multisectoral team led by the Camp Coordination

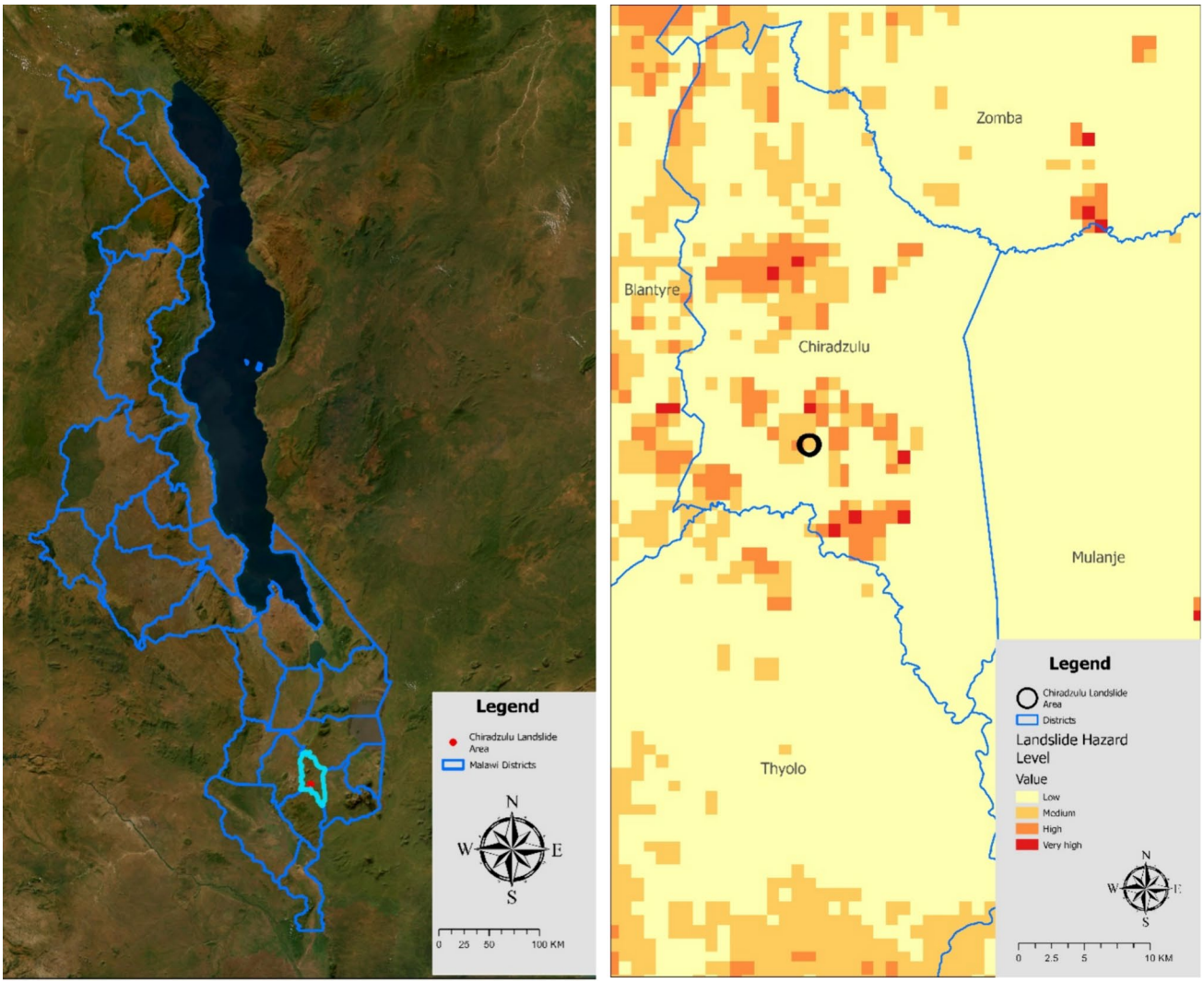


Fig. 1 Landslide hazard in Chiradzulu District (highlighted in the left image) in Malawi and the case study relocation village, Mtauchira Village (black circle in the right image)

and Camp Management (CCCM) sector, co-led by the International Organization for Migration (IOM), Shelter, WASH, and Protection sectors, including government counterparts came together and developed a Pilot Relocation Plan for Chiradzulu in July 2023 using the resettlement with dignity framework. The pilot project's primary goal was to support the Chiradzulu District to find long-term and safe solutions to end the temporary settlement of the displaced population. This included facilitating the request of funds, enabling coordination and engagement between national and district-level authorities and humanitarian partners, and facilitating stakeholder participation in an orderly phase-out of displacement sites.

The plan examined five major aspects of a dignified resettlement process including: settlements and shelter activities, water, sanitation, and hygiene (WASH), health, education, and protection (Table 1). Key activities (criticality, positive and negative aspects, budget, time frame, and technical skills required) were identified for each of these areas and collectively assessed. In addition, the plan also included priorities of each activity based on the input from variety of stakeholders taking part in plan preparation. Figure 3 shows the Location of Mtauchira village relative to the relocation site and a view of the relocation site.

4 Results

Using the materials and methods described in Sect. 3, we developed the Road to Resettlement Game. In this section, details of the game components and rules, game mechanics, and results of some test plays are presented.



Fig. 2 Drone images of Mtauchira Village after being impacted by the landslide, taken in August 2023 from the bottom of the slope (top left) and the top of the slope (top right), landslide impacted area of the village (bottom left), and the village with the location of destroyed buildings overlayed (bottom right)

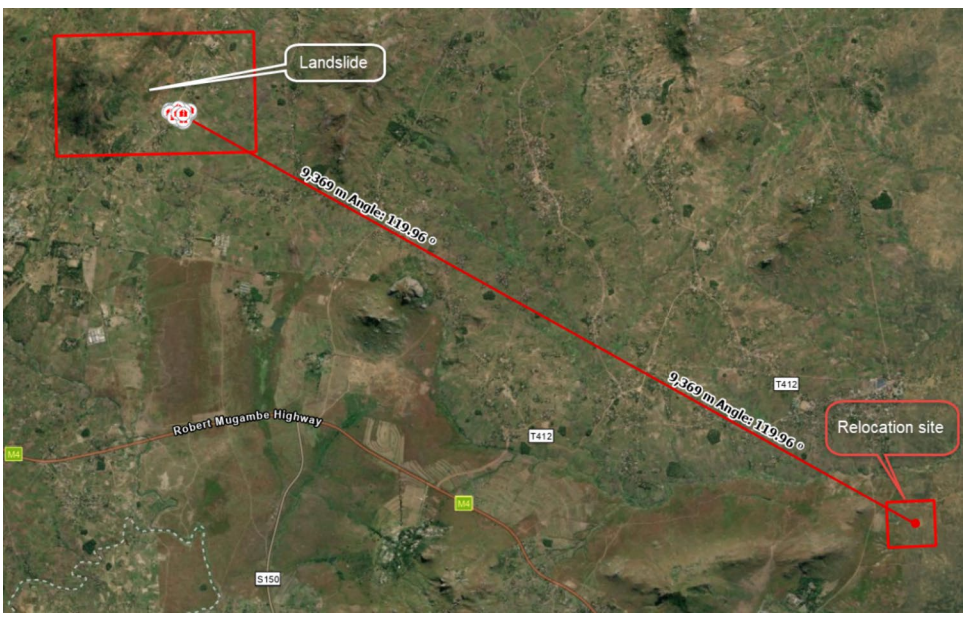


Fig. 3 Location of Mtauchira Village (top-left site) relative to the relocation site

Table 1 List of sectors and resettlement activities under each sector for the proposed resettlement project

Sector	Activities
Settlement and shelter	Geophysical assessment
	Environmental analysis on trees and crops that can be grown around the relocation site
	Road upgrading to gravel standard
	Installation of culverts
	Installation of water drainage
	Tree planting for soil stabilization
	Preparing development plan for housing construction
	Land survey and demarcation
	Land leveling
	Supporting the development of title deed
	Support for construction of houses
	Power and energy (energy saving stoves)
	Livelihood and employment opportunities
Water, sanitation, and hygiene (WASH)	Water quality at existing boreholes
	Rehabilitation of boreholes
	Drilling new boreholes (handpump)
	Finding and training of borehole mechanics
	Support for construction of latrines
	Constructing a solarized water supply system
	Enhancing water point operations and maintenance
	Conducting Knowledge, Attitudes, and Practices (KAPs) Survey
	Sanitary surveys
	Community consultation for latrine construction
	Identification and training of masons for latrine construction
	Activation and training of village health committees
Health	Disaster risk training and simulations
	Construction of outreach clinics
	Nutrition mass screening
	Risk communication and community engagement (RCCE)
	Indoor residual spraying to control vector borne diseases
	Mobile clinic services (MHPSS)
Education	Construction of junior primary school and three teacher's houses
	Children's corner in community and schools
	Planting woodlots and orchards in homes and communal areas
	Conducting life skill development for childcare protection
	Increasing promotional programs for boys and girls' education
Protection	Community engagement and sensitization
	Land ownership sensitization and administrative support
	Community sensitization on gender, child participation, inclusivity, and protection from sexual exploitation and abuse (PSEA)
	Community policing forum
	Strengthening community-based complaints and feedback: Health, education, safety
	Child protection committee
	Establishment of community-based childcare centers
	One stop center
	Community victim support unit
	Women and girls' safe spaces

Source IOM (2023)

4.1 Game Components and Rules

In the pursuit of combining the concepts of resettlement and relocation with the gameplay rules, the Road to Resettlement Game was designed with a progression divided into six primary levels: Land and Site Preparation, Housing and Livelihood, WASH, Health, Education, and Protection. The sequential completion of these levels by players is structured in accordance with the principles of resettlement (Tadgell et al. 2018). Figure 4 shows the table-top board game components consisting of the main map, community cards, game cards, and money cards.

Figure 5 shows an example of the community card and different activities that should be taken to complete each section of the game.

Players must complete the high priority activities of the "site preparation" and "housing and livelihood" sections sequentially as without them, a resettlement community could not exist. Each game starts with "Land and Site Preparation" that includes a variety of activities that must be completed in order to pass the level. Completing an activity means paying the required cost and having the human and technical capacity that are displayed on that activity card. As shown in Fig. 5, color codes have been used to categorize steps into three groups: high, medium, and low priority on the community cards. Red stand for high priority activities, yellow for medium priority activities, and green for low priority activities.

In order to pass each level, players must finish all high priority activities associated with that level. However, to complete the game, players must also eventually complete medium and low priority activities in the order of their choice. Figure 6 provides an illustration of a high priority card associated with activities in the initial level—Land and Site Preparation. While the front side of the cards remains the same, the reverse side shows the specific activity, cost, and technical personnel needed to execute the card.

The third essential element is the game map. As players successfully execute activity cards, they advance in their journey to complete the resettlement process. The player who successfully completes all 30 resettlement activities/projects/programs first and reaches the center of the board is the winner of the game. Figure 7 shows the map of the game and the route that each player is required to follow.



Fig. 4 Road to Resettlement Game components

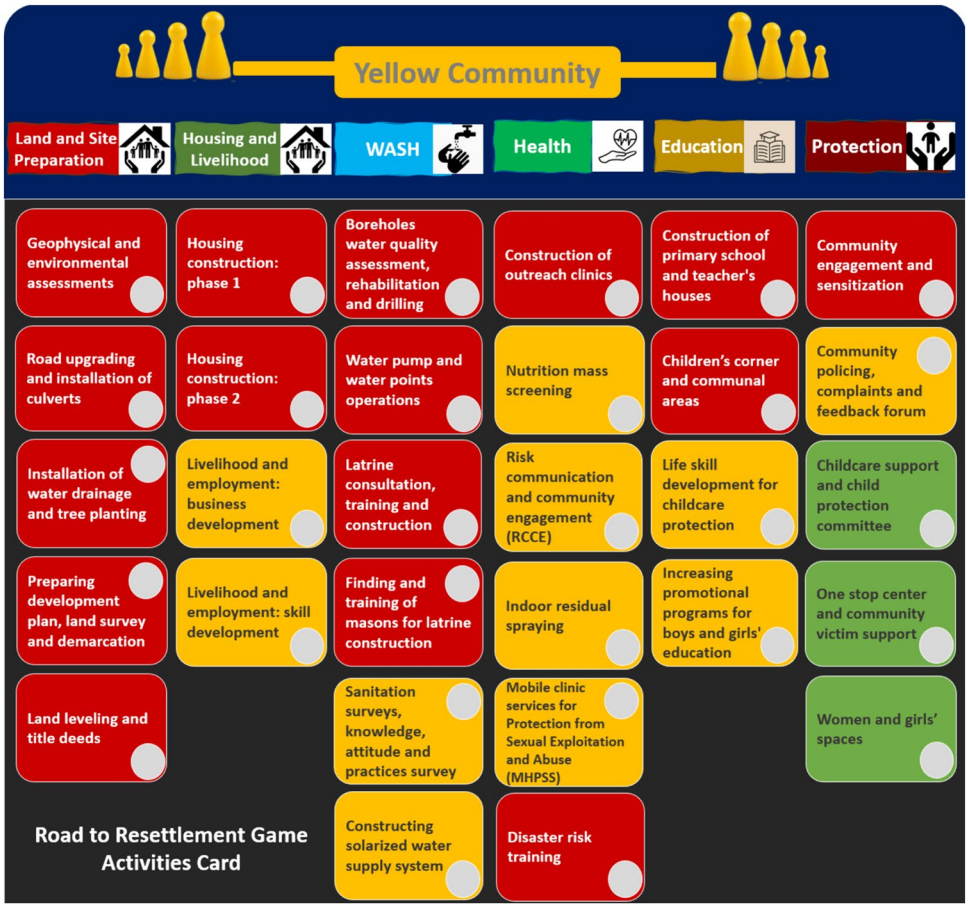


Fig. 5 Sample community card provided to each player. Each column represents a section of "safe and dignified resettlement" and its corresponding activities. Red, orange, and green colors show high, medium, and low priority. Completed activities by the player are checked marked as the game progresses

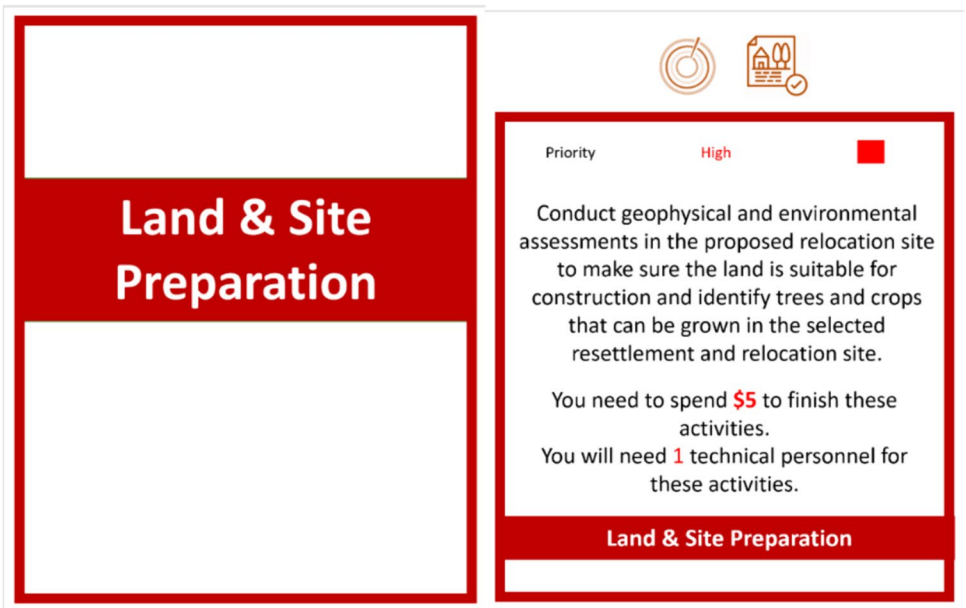


Fig. 6 A sample of the front and back sides of an activity card

4.2 Game Mechanics

In this game, four players participate, each selecting a colored pawn and positioning it in the start section (Fig. 7). Players then roll the dice, and the one with the highest number initiates the game. The initiation involves the first player drawing a card from the deck. As previously mentioned, players must first complete the Land and Site Preparation

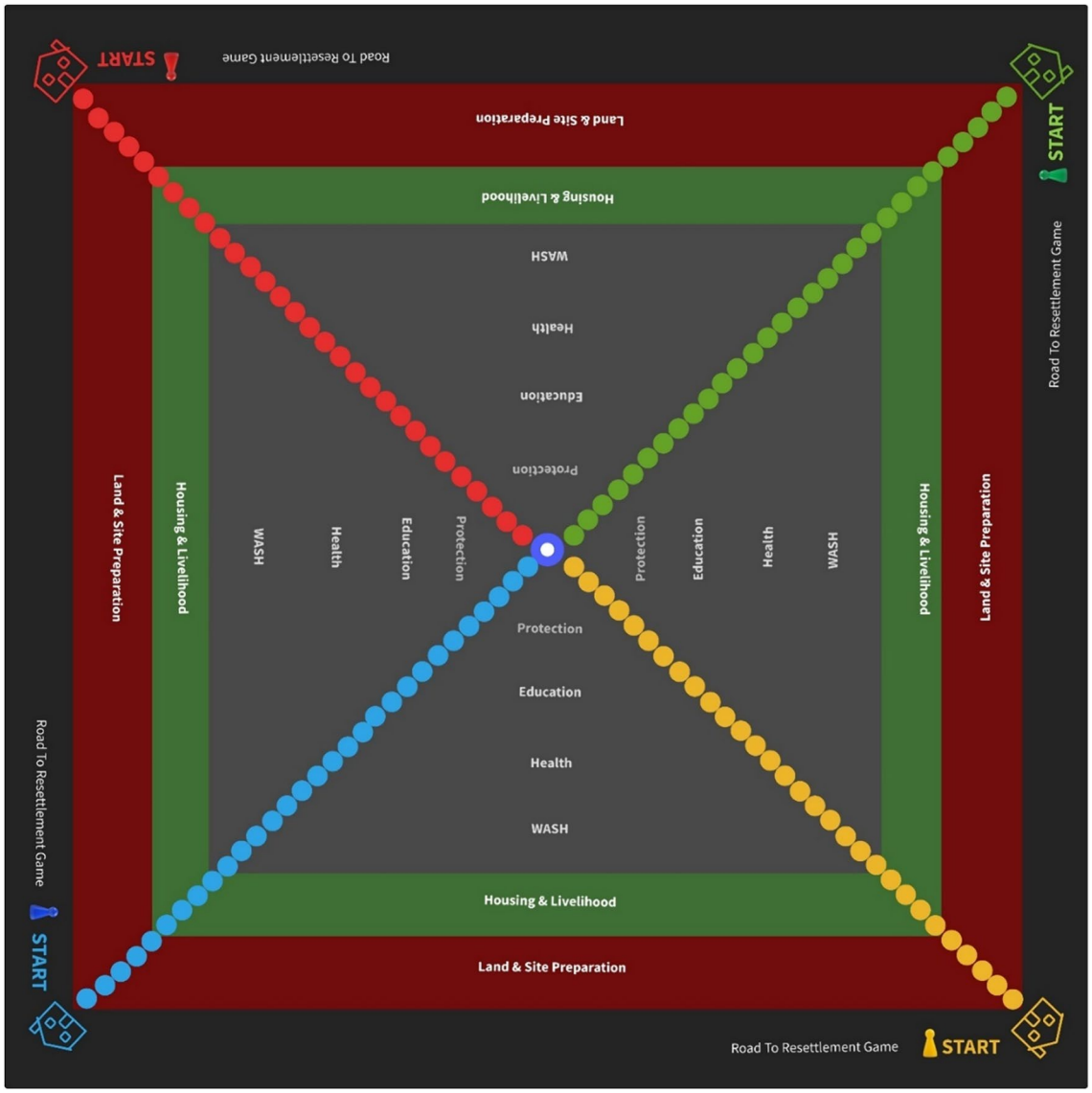


Fig. 7 The map of the road to resettlement game

level. If the drawn card pertains to this level, the player continues; otherwise, the turn passes to the next person.

For cards related to the first level, players read the description, pay the activity cost, and mark the white circle corresponding to that activity. Additionally, they advance one step along their designated path on the game map. Regarding budgeting, each player begins with $35. They use this budget to cover activity costs. When they run out or have an insufficient amount, they roll the dice for fundraising, gaining funds based on the dice number (1 = $10, 2 = $20,...,6 = $60). With the raised funds, players can implement the activity, moving their pawn one step on the board. Otherwise, they retain the card for the next round.

To introduce an element of uncertainty mirroring real-world scenarios, the deck includes several "Threats" cards. As the name suggests, these cards signify potential challenges in the resettlement process. When a player draws a Threats card, they must respond with the appropriate action and pay for that action if they have sufficient funds, either before or after fundraising. Figure 8 provides an example of a Threats card that players might encounter and need to address during the game.

This game mechanic offers multiple benefits to the players. First, it educates them on the sequential nature of the resettlement process, emphasizing that certain aspects, such as health, cannot be addressed without prior attention to housing and livelihood. This sequential understanding helps players avoid confusion in critical circumstances. Moreover, the incorporation of budgeting considerations and the management of threats exposes them to various dimensions



Fig. 8 Front and reverse side of a Threats card

of a real resettlement process, familiarizing them with the complexities they might encounter.

4.3 Test Plays

The Road to Resettlement Game underwent multiple test plays with experts and scholars to optimize its functionality and player engagement. In the initial playtests, it became apparent that players were overly fixated on their budgets and the costs of activities, which compromised the educational goals of the game. To address this, we increased the font size of the activity card descriptions and reduced the size of cost numbers. This visual adjustment steered players towards focusing on the activities necessary for progressing in the game. Figure 9 shows a couple of test play of the game.

Moreover, in order to enhance the educational aspect of the game dynamics, we introduced a priority feature to activity cards. This addition provided players with the opportunity to strategically allocate their budget by implementing high-priority activities first. Balancing the completion of all levels while making resource-conscious decisions aligns with the second significant purpose of the game, which is to impart knowledge and ensure engagement.

5 Discussion

The Road to Resettlement Game significantly emphasizes the crucial element of community participation that is one of the most important requirements for successful resettlement. Safe and dignified resettlement framework used in developing the game aligns well with the established studies that advocate for the engagement of both affected and unaffected communities in effective disaster risk reduction, especially during the recovery and reconstruction phases (Pearce 2003). Playing the game, participants assume the roles of decision makers, such as government officials or community leaders, and navigate through diverse scenarios, considering factors such as infrastructure, community needs, human and financial resources, and environmental sustainability. This immersive experience empowers players



Fig. 9 Sample test plays of the road to resettlement game

to orchestrate community involvement during a realistic situation, encompassing a comprehensive list of activities that should be implemented sequentially. Additionally, engaging in the game enhances players' awareness and knowledge of potential current and future threats throughout the resettlement process. Playing such game in advance of an actual project will enable participants to be better informed about future challenges and obstacles and take actions to address them. As such it can help filling some of the existing knowledge and critical decision-making gaps in post-disaster resettlement projects (Balachandran et al. 2022).

Furthermore, the game incorporates a crucial element by differentiating between various activities based on their priority and significance. This classification was determined by experts who systematically compiled and prioritized all essential components necessary for a successful, safe, and dignified resettlement process. By categorizing these actions, the game provides a framework that enables those responsible for implementing resettlement to make informed decisions, emphasizing the importance of high-priority actions in case of resource or time constraints. This approach allows players to revisit and address these prioritized activities at a later stage, ensuring that their completion does not pose significant issues even if initially omitted. Such ability to prioritize among different actions and making trade offs are key to serious games in all contexts particularly in socioenvironmental issues (Neset et al. 2020).

The other crucial factor of the game is the inclusion of "Threat" cards. These cards add nuances to the narrative of the game including local customary and geographical specificities. While grasping the fundamental principles of storytelling is essential for comprehending and crafting narratives for serious games, the interactive nature of games introduces additional challenges. Even when the story remains unchanged through interaction, the player's actions must be considered in presenting the narrative (Kampa et al. 2016). Threats cards also dramatize the game. For example, when a player is only focusing on the activity cards and believes they can win the game with the remaining budget they have, the appearance of a threat card disrupts their strategy and forces them to devise new ones.

Finally, in the process of preparing activity/project/program cards, we tried to include all essential elements of the safe and dignified resettlement framework. This framework considers essential infrastructure needed for resettlement (such as road upgrading and culvert installation, development plan preparation, land survey and demarcation, housing construction, and operation of water pump and water points). It also considers other aspects such as community involvement (as represented by cards like Risk Communication and Community Engagement, Construction of Primary School and Teacher's Houses, and Livelihood and Employment) and minority and vulnerable groups as represented by cards like Children's Corner and Communal Areas, Women and Girls' Spaces, and Childcare Support and Child Protection Committee. This incorporation into activity cards makes the game a comprehensive resettlement tool that elevates the level of organization in the resettlement process during high-stress times. Lack of attention to these aspects in resettlement projects are among the well documented issues (Balachandran et al. 2022).

There are few limitations in this game that may cause restrictions from generalizing it for all communities and players. Although the game is functional for a wide range of students, practitioners, and active community members interested or involved in resettlement, it would be more educational and effective if players have some background knowledge and understanding of the terms used in the game. It is also important to note that this game was developed based on the collective opinions of the experts who participated in the sample resettlement project in Malawi. While these values may work for the learning purposes of this serious game, they may not be used as the basis for real-world decision making.

6 Conclusion

Serious games are increasingly being used in teaching, training, and community engagement. They can be used to transfer critical knowledge and skills to intended audiences from academia, humanitarian, and development sectors and governments in a short period of time. Considering the growing number of natural hazards events and population displacement and need for resettlement, we developed a serious game to help players appreciate the key processes, activities, challenges, and resources associated with resettlement and relocation. Built based on a real resettlement project in southern Malawi, the Road to Resettlement Game components mimic the real-world activities and concerns that need to be resolved in each resettlement phase to achieve a safe and dignified resettlement.

The Road to Resettlement Game can be used by practitioners and local officials who will be involved in post-disaster resettlement and relocation projects as a learning tool to understand and appreciate the complexity and systemic nature of these projects. Students of relevant disciplines (architecture, planning, public health, humanitarian and emergency management, social work, nursing, civil engineering) and practitioners can use this game to learn how resettlement projects in post-disaster situations work. The Road to Resettlement Game is incredibly low-cost, making it accessible to all users for training and instructional purposes. To make the game more widely available, we are aiming to create an online version of it.

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